Venkata Lakshmi Parimala Pasupuleti

Backend Engineer · Agentic AI Evaluation · LangGraph · LangChain · MCP · Observability · RAG Pipelines · Python · PropTech · 9yr Production

+1-703-261-5410 | venkatalakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

Backend engineer with 9+ years of production experience and a deep specialization in agentic AI evaluation, tracing, and observability - the exact platform Zillow's Agentic Evals team is building. AgentProbe is the direct proof point: a production evaluation framework I designed and shipped end-to-end with a 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency), LLM-as-Judge scoring, pass@k multi-trial harness surfacing variance in non-deterministic outputs, CI-integrated regression gate blocking drift before deployment, offline batch evaluation pipelines, online scoring with cost and latency tradeoffs, and MCP integration for AI-to-system data retrieval - the same evaluation framework, tracing system, and feedback loop infrastructure Zillow's Evals platform provides to domain teams. The platform is open-source, multi-team deployable, and built with the same developer-facing API and internal customer mindset this role requires. I build with LangGraph, LangChain, and MCP daily - all three frameworks named in this role - and use Claude Code as a core part of my engineering workflow. LeaseI demonstrates the multi-agent production system the Evals platform must instrument and measure: 8-agent LangGraph StateGraph with conditional routing, shared memory, tool invocation, and HITL guardrails. At Entrata (real estate/PropTech SaaS), I owned production systems for 7 years at 99.5% uptime - the reliability and scalability the evaluation layer requires. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Agentic Evals & Observability: Python (9yr), LangGraph, LangChain, MCP (Model Context Protocol), Evaluation Frameworks, Offline Batch Evals, Online Scoring, LLM-as-Judge, Hallucination Detection, Groundedness, pass@k Harness, Cost/Latency Tradeoffs, Tracing Systems, Observability, Feedback Loops, CI Regression Gates, Drift Detection, Quality Measurement, Prompt Engineering, Prompt Versioning, Anthropic API, OpenAI API, Ollama, Claude Code

Platform & Infrastructure: Multi-Agent Systems, Agent Orchestration, Tool Calling, RAG Pipelines, ChromaDB, Vector Databases, Semantic Search, FastAPI, React.js, TypeScript, REST APIs, API Design, Developer-Facing APIs, Platform Infrastructure, Scalable Systems, AWS (S3, EC2, RDS, Glue, Lambda, DynamoDB, Bedrock), Docker, Microservices, GitHub Actions CI/CD, Automated Testing

Engineering & Domain: Distributed Systems, Reliability Engineering, Incident Response, Logging, Runbooks, Performance Engineering, Technical Mentoring, Code Reviews, Design Docs, PostgreSQL, MySQL, Oracle, Redis, RabbitMQ, Linux, Git, Agile/Scrum, PropTech Domain, Real Estate SaaS

PROJECTS

AgentProbe: Production Evaluation Framework - Agentic AI Systems

Evaluation Framework · Tracing · Observability · CI Gate · Open Source

Python · FastAPI · LangGraph · ChromaDB · MCP · GitHub Actions · AWS Bedrock-compatible · Anthropic API · OpenAI API | github.com/venkatalakshmiparimala/AgentProbe

Production evaluation framework and tracing platform - the exact system Zillow's Agentic Evals team builds for domain teams. **Evaluation engine:** 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency); LLM-as-Judge scoring; pass@k multi-trial harness surfacing variance in non-deterministic outputs; online scoring with cost and latency tradeoffs. **Observability & feedback loops:** tracing systems with structured logging across agent steps; CI-integrated regression gate (GitHub Actions) blocks quality drift before deployment - making evaluation a first-class engineering practice. **Offline batch evaluation:** pipeline runs evaluation suites at scale without live system load. **Platform:** developer-facing API; multi-provider LLM routing (Anthropic, OpenAI, Ollama, Bedrock-compatible); MCP integration; 15-endpoint FastAPI backend. Open-source, multi-team deployable. Zero regressions across 1,240+ test cases.

LeaseI: Multi-Agent Production System - What the Evals Platform Must Instrument

Multi-Agent · LangGraph · HITL · Non-Deterministic · Production

Python · LangGraph (8-agent) · FastAPI · RAG · SQLite · MCP · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production multi-agent system that demonstrates exactly what the Agentic Evals platform must instrument and measure. **Agent orchestration:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Escalation) with conditional routing and dynamic task delegation based on confidence scores - the non-deterministic routing behavior that evaluation harnesses must handle. **Shared context:** SQLite knowledge repository persisted across all agents and sessions - coordination layer enabling parallel agent workflows. **Tool invocation:** agents invoke retrieval, structured data, and configuration tools with grounded, citation-backed outputs. **Guardrails:** deterministic HITL escalation blocks unsafe decisions. 93.6% precision, 87.9% F1 across 1,240 test cases - metrics the evaluation framework tracks in production.

AI on FHIR: Full-Stack Agentic System - Healthcare API Integration

Full Stack · React · AWS · Tool Use · HIPAA

Python · Flask · React.js · TypeScript · Docker · AWS · BERT · spaCy · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack agentic system with React frontend, Python backend, and AWS infrastructure - LLM-powered tool invocation translates natural-language queries into structured API calls against FHIR healthcare data; HIPAA-compliant RBAC with authentication and full data isolation; complete audit trail enabling evaluation traceability; ~60% reduction in manual retrieval time. Demonstrates full-stack ownership (React through backend through scalable cloud infrastructure) and production observability patterns for regulated agentic systems.

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Production Agentic AI Delivery

- Designed and deployed a production agentic AI system (LangGraph, FastAPI, Python) for an external customer - sole technical owner from requirements through architecture, implementation, evaluation pipeline, and production deployment; wrote design docs; built HITL guardrails and validation gates for regulated decisions; delivered tracing, observability, and feedback loop infrastructure for ongoing operation.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977); designed observability dashboards; mentored 600+ students in Python, SQL, and distributed systems; delivered complex technical outputs for non-technical stakeholders.

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Production systems at PropTech scale** - owned HUD compliance pipeline (180,000 domain records, 100K+ housing units, real estate industry) for 7 years at 99.5% uptime; Docker microservices architecture redesign (50% scalability improvement, 30% deployment time reduction); RabbitMQ event-driven async with retry logic, idempotency guarantees, and dead-letter handling; AWS S3 document migration (100% PHI compliance, zero data loss) - the reliability and scalable infrastructure the evaluation layer requires.
- **Platform engineering and observability** - Python, FastAPI/Flask REST APIs, React.js, TypeScript, AWS S3/EC2/RDS/Glue, Docker, PostgreSQL/MySQL/Oracle, Redis; meaningful alarms, runbooks, and incident response with RCA; 200+ SQL optimizations (30% performance gain); internal debugging tools reducing triage time 40%; GitHub Actions CI/CD with automated testing; bug count below 10 for 6+ consecutive months across 20+ production modules in 120+ Agile sprints.
- **Technical leadership** - led projects spanning multiple engineers; wrote design docs and architecture decisions; mentored 5+ engineers on system design and production practices; 300+ peer code reviews; Python/Bash automation eliminating 80% of manual workload; cross-functional delivery with product, compliance, and operations stakeholders.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (Python, PostgreSQL) for production SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017